

Po-Sheng Cheng

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bencer@outlook.com

Portfolio: <https://bencer3283.github.io/art/>

Master of Industrial Design, Rhode Island School of Design, exp. Jun. 2026
Cross-enrolled at School of Engineering, Brown University
B.Sc. Bio-Mechatronics Engineering, National Taiwan University, Jan 2023

Skills

- Coding: C/C++, JavaScript (React, Express, MongoDB), Dart (Flutter), Python (PyTorch, NumPy), Git, Linux
- Electronics: PCB design (Altium), STM32, ESP32, Raspberry Pi, system interface (I2C, SPI, UART)
- Mechanical: CAD (SolidWorks, Fusion, CFD and Creo), rapid prototyping (FDM/SLA 3D Printing)
- Design: Rhino, Keyshot, Figma (Wireframing), Adobe Suite

Experiences

Product Development Apprentice, Google, Jun. - Sep. 2024

Taipei, Taiwan

- Led a team of 5 to design a Hardware-as-a-Service platform for managing lost items in public facilities. Laid out the system architecture of the hardware kiosk, user-facing app and web services.
- My prototypes of Raspberry Pi and ESP32 hardware, a React-based website, a mobile app and a RESTful API service drove the development direction from a single hardware product to an integrated platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022

Hsinchu, Taiwan

- Created a brand new breed of keyboard technology by conducting quantitative user research with 50 interviewees and innovating an Arduino microcontroller-based prototype to control actuators like stepper motor and electromagnet. Also lots of CAD (Creo) and SLA/FDM prototyping on the mechanical design.
- The prototype received widespread praise across many departments (pm, engineering, design) in the company while I worked with them to support guiding the direction of product development.

College Student Researcher, NTU, Jul. 2021 - Feb. 2022

Taipei, Taiwan

- Developed a novel, integrated spectral mapping system that vastly accelerates scientific spectral measurement.
- By using linear dispersion optics and optimizing the EMCCD readout algorithm in LabVIEW and C++, the scanning time was reduced by 26%.
- Awarded the **2021 Technology Innovation Award** by CCMS, NTU and **College Student Research Creativity Award** by National Science and Technology Council of Taiwan with this project.

Projects

MITM Router

- Built a WIFI router that brings physical, tactile interactions to the user's internet usage with Linux networking stack on a Raspberry Pi.
- Extensive prototyping and user testing of the physical user interface was done using several sensors/actuators on a RP2040 microcontroller.

PageMate smart bookmark

- Built the physical device with advanced touch gesture sensing on ESP32 and the Node.js (Express) web service of a smart bookmark that can display reading progress of the user's ebooks.
- Used Figma to create a demo experience that can be updated in real time using my Node.js web service.

(Publication) TeleShift: Teleexisting TUI for Physical Collaboration & Interaction

- Utilized ESP32 to implement a shape-transforming device with a 3D tangible user interface that enabled this work to be awarded **Best Demo** at 2022 ACM Ubicomp conference.
- Designed the robust interactive mechanisms (SolidWorks). My design for manufacturability (DFM) improvements also slashed assembling time by 80% during demo.

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Portfolio Highlights

Hardware Projects

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MITM Router

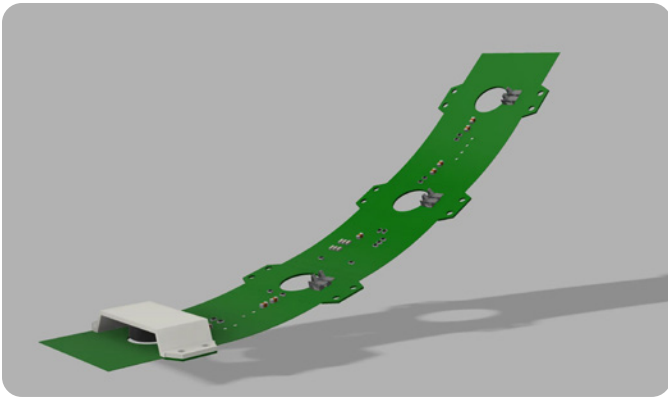
an interactive WI-FI router



Raspberry Pi 5 running Linux networking stack
ESP32 and RP2040 microcontroller
7+ peripheral sensors, actuators or ICs

Haptic Belt

using haptic feedback to provide navigation for visually impaired people



Rigid-flex PCB design with Altium Designer
Power integrity analysis
STM32 firmware programming ([Github Repo](#))
RF circuit design

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ESP32 microcontroller
Extensive use of I2C
Custom IC for touch gesture sensing

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Portfolio Highlights

Software Projects

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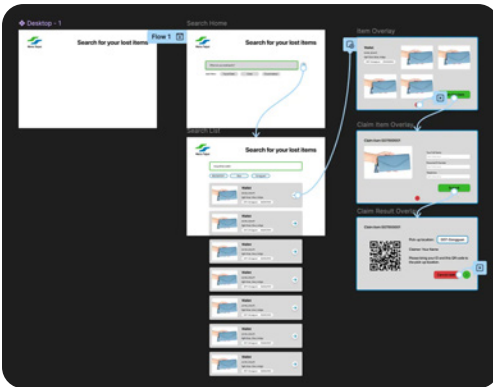
My Online Photography Gallery



Built with: Flutter/Dart, Netlify,
Google Drive API
[Design system](#)
[GitHub Repo](#)

Nexus

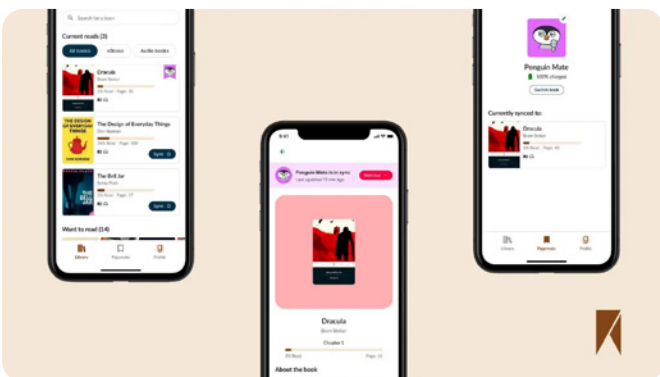
intelligent lost item recognition and management platform



Built with: Flutter/Dart, React,
Figma, Linux on Raspberry Pi
[Figma prototype](#)
[GitHub Repo](#)

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Built with: Node.js (Express), Figma